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Comparing static versus dynamic protected areas in the Québec boreal forest

Bronwyn Rayfield^{a,*}, Patrick M.A. James^b, Andrew Fall^c, Marie-Josée Fortin^a

^aDepartment of Ecology and Evolutionary Biology, University of Toronto, 25 Harbord Street, Toronto, Ontario, Canada M5S 3G5

^bFaculty of Forestry, University of Toronto, 33 Willcocks Street, Toronto, Ontario, Canada M5S 3B3

^cGowlland Technologies Limited, 220 Old Mossy Road, Victoria, British Columbia, Canada V9E 2A3

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ABSTRACT

Conservation planning is often based on static mapping of species' ranges or habitat distributions. Succession and disturbance alter, however, habitat quality and quantity through time especially under global climate and land use change scenarios; hence, static protected areas may not ensure habitat persistence and species survival. Here, we examined the relative merits of static and dynamic (floating) protected areas for the conservation of American marten (*Martes americana*) habitat in a dynamic boreal forest of Québec (Canada). Forest dynamics were modeled using a spatially-explicit landscape disturbance model and protected areas were selected based on the quality and compactness of marten home ranges using MARXAN. Static protected areas were fixed in space during 200 year simulations of boreal forest dynamics, while dynamic protected areas were re-located every 50 years to track dynamic habitat. Dynamic protected areas supported more high quality home ranges through time than static protected areas. The locations of dynamic protected areas were constrained, however, by the highly fragmented forest patterns created through logging and fire in unprotected areas. Our findings emphasize the often-overlooked point that if dynamic conservation planning is to be successful in the long term, the landscape matrix quality surrounding protected areas must be managed in such a way that options remain when it comes to re-planning.

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1. Introduction

Protected areas play a vital role in safeguarding against further losses of existing biodiversity and wildlife habitat that result from increased human encroachment on natural spaces (Williams et al., 2004). To fulfill this role, protected areas (hereafter PAs) must not only ensure that multiple elements of biodiversity are adequately represented but that they persist in their protective role through time in dynamic landscapes (Margules and Pressey, 2000; Gaston et al., 2006). In the last two decades, numerous site-selection algorithms

have been developed to identify sites to be included in one or more PAs that maximize the amount of extant biodiversity represented given existing economic constraints. These algorithms typically rely on static distribution patterns of biodiversity surrogates such as species (Csuti et al., 1997), habitat types (Olson and Dinerstein, 1998), and land types (Pressey et al., 1996) to identify, prioritize, and select candidate PAs. However, because distribution patterns of species and habitats change over time due to natural processes (e.g. population dynamics, succession, and disturbances) and human impacts (e.g. land use and climate change), site-selection

* Corresponding author. Tel.: +1 416 946 7886.

E-mail addresses: bronwynrayfield@zoo.utoronto.ca (B. Rayfield), patrick.james@utoronto.ca (P.M.A. James), andrew@gowlland.ca (A. Fall), mjfortin@zoo.utoronto.ca (M.-J. Fortin).

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algorithms based on representation objectives alone do not necessarily select PAs that ensure the long-term biodiversity persistence (Margules et al., 1994; Rodrigues et al., 2000).

Promoting long-term persistence of biodiversity in PAs therefore requires that site-selection algorithms move beyond simply meeting initial representation targets towards incorporating information about the spatio-temporal dynamics of populations and landscapes (Cabeza et al., 2004). Efforts to include population dynamics into site-selection algorithms have treated species persistence objectives either implicitly by prioritizing higher quality habitat (Williams and Araujo, 2000) in large, compact, and well-connected PAs (McDonnell et al., 2002) or explicitly by modeling species-specific persistence through population viability analyses (Carroll et al., 2003) or metapopulation models (Moilanen and Cabeza, 2002). Landscape dynamics have only recently been incorporated into site-selection algorithms and have primarily taken the form of an implementation problem in which PAs must be implemented gradually over several time periods due to budget constraints (Pressey and Taffs, 2001; Meir et al., 2004; Oetting et al., 2006). These studies incorporate the possibility that unprotected sites may be lost or degraded before the next implementation period due to anthropogenic landscape changes such as development or resource extraction. The conservation implications of stochastic landscape processes such as succession, natural disturbance regimes, and climate shifts that operate on timescales longer than the typical PA implementation horizon have just begun to enter discussions about PA design and have remained fairly theoretical (Cumming et al., 1996; Bengtsson et al., 2003; Araujo et al., 2004; Pyke et al., 2005; Williams et al., 2005; but see Pickett and Thompson, 1978). A few studies have proposed, however, methods to incorporate dynamic ecological processes that provide important ecosystem services such as specialist pollinator interactions, hydrological regimes, trophic interactions, and ecological diversification of species lineages into quantitative conservation targets (Pressey et al., 2003; Chan et al., 2006).

The Québec boreal forest is a good system to examine the implications of landscape dynamics for conservation planning because it is perpetually dynamic due to both stochastic natural disturbances and deterministic human activities. Wildfire is the dominant natural disturbance agent and is highly variable in terms of extent, frequency, and ecological impact (Johnson, 1992). Logging, the dominant human disturbance agent, eliminates or substantially reduces the mature portion of these forested landscapes (Bergeron et al., 1999), and this has motivated the creation of a system of PAs to protect mature forest habitat and act as constraints to the logging schedule (Baskent and Keles, 2005). Selecting PAs in landscapes driven by highly variable disturbance regimes may be difficult because representative areas may not exist even at very broad scales (i.e. there may not be a smaller part of the region that is similar to the whole region; Cumming et al., 1996). The lack of representational areas in boreal forests in terms of an appropriate compositional measure has led some researchers to propose an alternative to static PAs, in which representation and persistence criteria are re-assessed periodically through time to create a dynamic system of PAs that change locations to track dynamic habitat (Cum-

ming et al., 1996; Bengtsson et al., 2003). Dynamic protected areas may be able to overcome some of the difficulties presented by highly variable natural systems when planning for conservation by ensuring that adequate habitat is always available. An intuitive illustration of the dynamic PA concept occurred as early as 1983 in the Gifford Pinchot National Forest when a windstorm damaged nesting sites in six out of forty-five PAs for spotted owls (*Strix occidentalis*) resulting in the re-location of one PA and the merging of two others (Ruediger, 1985).

When considering the implementation of dynamic PAs, an important distinction must be made between PAs implemented over broad, regional scales (Carroll et al., 2003) and those PAs implemented within multi-use forest landscapes subject to logging (Baskent and Keles, 2005). The scale of PAs will determine which criteria and processes should be considered during the selection procedure. At a regional-scale, PAs may be large enough that they can maintain internal re-colonization sources under the natural disturbance regime (Pickett and Thompson, 1978); hence, dynamic PAs at this scale are unnecessary because species' occurrences alone in these PAs should serve as a surrogate for their long-term persistence. Smaller PAs implemented at local scales may also play an important role in biodiversity protection by preserving species or habitats that are under-represented in regional-scale PAs, providing a source of propagules and offspring for re-colonization of logged areas, and facilitating the movement of biota across managed landscapes (Lindenmayer et al., 2006). Species persistence in these local scale PAs will be highly dependent on landscape dynamics as well as spatio-temporal population dynamics; hence allowing these PAs to be re-located in response to changes in habitat quality and configuration may improve their ability to conserve target species.

Although several conceptual papers have suggested the use of dynamic PAs in boreal forest ecosystems (Cumming et al., 1996; Bengtsson et al., 2003; Angelstam et al., 2004), none have discussed the details of their implementation. In this study, we investigate the potential of using dynamic rather than static PAs to maintain old growth habitat within the boreal forest and explicitly describe the steps necessary to identify, prioritize, and select PAs through time. The total area of PAs was fixed between 10% and 12% of the study area to allow for alternative forest uses such as timber harvest and to comply with conventional conservation targets (McNeely, 1993; IUCN, 1994). Patches in the static PAs were fixed in space throughout the course of a 200-year simulation of boreal forest dynamics, while patches in the dynamic PAs were periodically re-evaluated and re-located in response to changes in forest conditions due to fire, succession, and logging. The length of the re-planning interval for dynamic PAs was assumed to be constrained by operational costs associated with PA re-location and was therefore conservatively set as 50 years corresponding to half of the harvesting age (100 years). PAs were chosen based on habitat requirements of the American marten (*Martes americana* Rhoads), which is often used as an indicator species for mature boreal forest biodiversity (Thompson, 1991; Buskirk and Powell, 1994) because of its large home range size (maximum home range size of 1000 ha; Chapin et al., 1998; Potvin et al., 2000) and its degree

of habitat specialization (prefers coniferous/mixed forest greater than 30 years old; Buskirk and Powell, 1994; Thompson, 1994). To select PAs, we first identified all potential marten home ranges in our study area-based on the spatial configuration of forest types and ages (James et al., 2005). We then used simulated annealing to find two spatially compact patches of high quality potential marten home ranges to form a PA (Possingham et al., 2000). Boreal forest dynamics were simulated using an existing spatially-explicit landscape-scale model (Fall et al., 2004) that includes sub-models for forest succession, fire, and logging. We hypothesized that dynamic PAs would outperform static PAs because the marten home ranges within the static PAs may deteriorate through time due to losses from wildfire. To compare static and dynamic PAs, we assessed the number, quality, and connectivity of potential marten home ranges remaining within PAs and across the entire landscape every 50 years over the course of our 200 year simulations.

2. Methods

2.1. Study area

The Vermillion study area is a boreal mixed-wood forest management unit that covers approximately 430,000 ha in south-central Québec (Fig. 1). The northern portion of the study area is a coniferous boreal forest characterized by black spruce (*Picea mariana* (Mill.) BSP.), balsam fir (*Abies balsamea* (L.) Mill.), jack pine (*Pinus banksiana* Lamb.) and white birch (*Betula papyrifera* Marsh.). The southern portion is a transition zone between the boreal and temperate forest regions characterized by white spruce (*Picea glauca* (Monech) Voss), balsam fir, and yellow birch (*Betula alleghaniensis* Britt.). Lakes and wetlands are distributed throughout the study area to make up approximately 10% of the total area. Stand replacing fire is the dominant natural disturbance (Hély et al., 2000) and logging has been the dominant anthropogenic disturbance since the 1950s. Data for the Vermillion study were derived from the 1990 decadal forest inventory carried out by *Système d'Information FOREstière par Tesselle* (SIFORT) in collaboration with the Québec Ministry of Natural Resources and Abitibi Consolidated. Spatial data of tree age and species composition were derived from supervised classification of aerial photography and are represented in raster format at a resolution of 0.25 ha (50 × 50 m).

2.2. Boreal forest landscape dynamics model

We used a spatially-explicit stochastic raster simulation model that integrates succession, growth, yield, and fire regime information to simulate landscape-scale forest dynamics. The model was implemented in SELES modeling platform (Fall and Fall, 2001) and consisted of sub-models for fire, succession, and logging that interact by modifying the dynamic forest conditions within each 0.25 ha pixel. Fall et al. (2004), Didion et al. (2007), and James et al. (2007) used this model to investigate interactions among forest management strategies and fire. We used this model to simulate boreal forest dynamics over 200 years with 10 replicates for both the static

and dynamic PA scenarios. The number of replicates was limited by the data processing time required to move between the simulation model and PA selection software.

2.2.1. Description of sub-models

Fire was modeled as a stochastic process based on empirical estimates of mean fire size and fire return interval (Van Wagner, 1978). The mean number of fires per year was a function of the size of the study area, the mean fire size, and the fire return interval.

Mean Number Fires per Year

$$= \text{Study Area Size} / [(\text{Mean Fire Size})(\text{Fire Return Interval})]$$

Fire sizes were selected randomly from an exponential distribution with a mean of 1500 ha (Fall et al., 2004). The exponential model is a commonly made assumption for studies of fire in the boreal and boreal mixed-wood forest and necessarily assumes independence among fire events (Van Wagner, 1978; Johnson and Van Wagner, 1985; Bergeron et al., 2002; Fall et al., 2004). Fire return interval, the number of years required to burn an area of size comparable to our study area, was estimated to be 250 years to reflect recent changes in climate (Bergeron et al., 2001). Fire ignition occurred randomly and fire events spread randomly from the point of ignition to any eight neighboring cells until the total area burned reached the pre-selected size. In this statistical fire model, fires burned independent of forest age or species composition so that the effects of different PA scenarios were not confounded by uncertainty in the burn probabilities for different fuel categories. This simplifying assumption emphasizes the effects of stand replacing fires in boreal forests which are strongly linked to extreme weather events during which fires burn young and old stands alike (Johnson et al., 2001; Lefort et al., 2003). Such fires account for more than 97% of the area burned although they account for less than 5% of the total number of fires (Stocks et al., 2002).

Forest succession was modeled using probabilistic successional pathways derived as a function of site-specific soil and drainage conditions and previous vegetation types based on long-term, plot-level data on forest transitions in the Vermillion study area (Fall et al., 2004). Forest stand age was incremented annually until a maximum of 300 years unless either a fire or logging event occurred, which reset stand age to zero. Forest stand composition likewise followed a probabilistic successional trajectory unless reset by a fire or logging event. In the absence of disturbance, cells continue to age to a maximum of 300 years and site-specific maximum volume. They remain in the maximum age and volume categories until they are either burned or logged, implicitly assuming that such stands are in a gap-phase old forest state.

To simulate the effects of logging, the logging sub-model used an area-based annual allowable cut (AAC) with a minimum harvest age of 100 years and site-specific road accessibility that reflected plausible management scenarios for the mixed-wood boreal region in Quebec. The AAC was set at 3900 ha (1% of the productive forested study area; Fall et al., 2004; Bergeron et al., 2006; Didion et al., 2007) and volume was estimated for each cell based on yield curves from plot data, but the details are not relevant here because our AAC was based on area not volume. Using an area-based logging

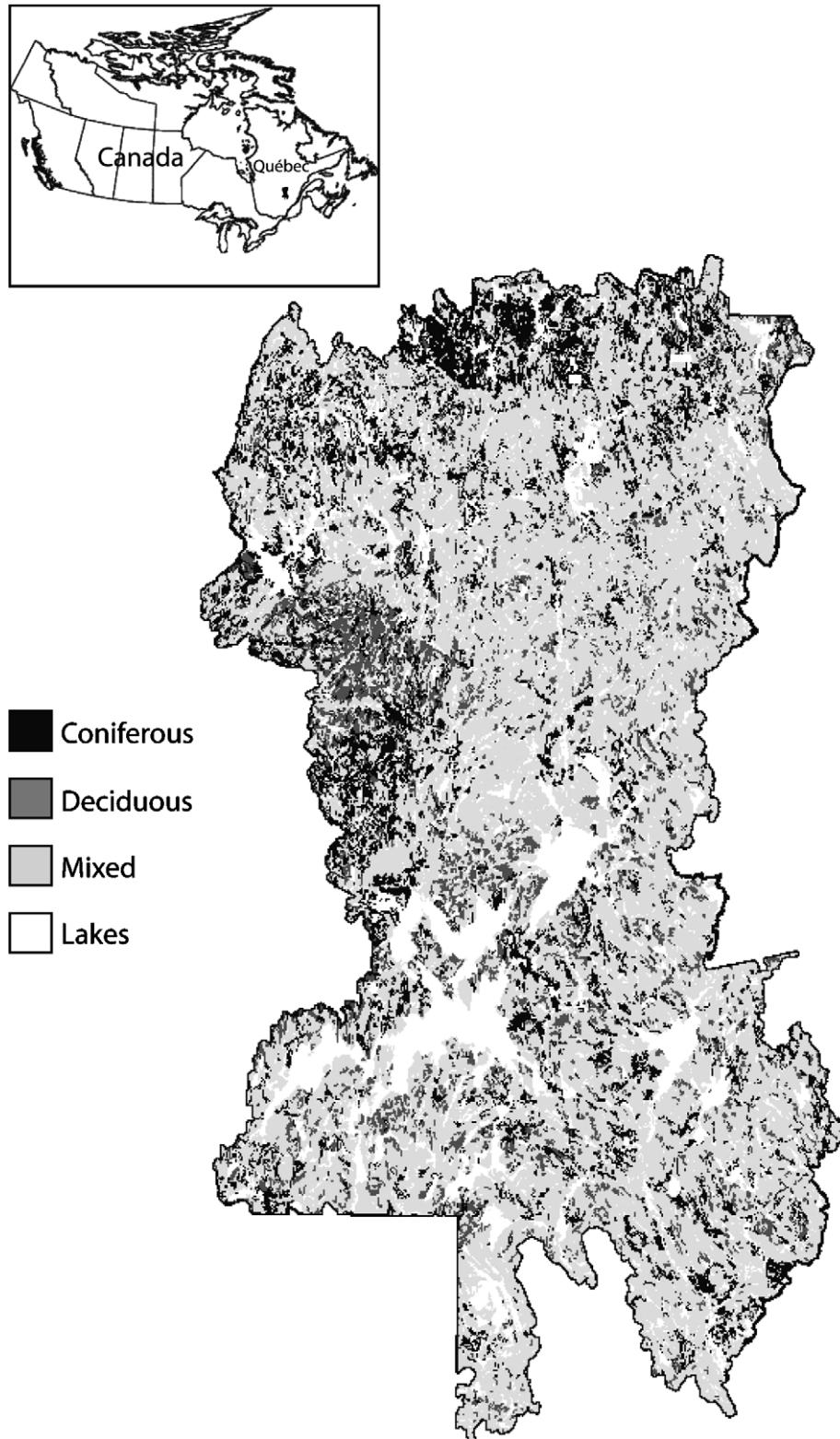


Fig. 1 – Location of the Vermillion study area in the province of Québec, Canada. The map of the study area shows the spatial arrangement of habitat types at a resolution of 0.25 ha/cell.

target allowed for greater control and transparency in the forest management process by providing a common currency (i.e. forest area) for the logging and conservation targets and the fire sub-model. Harvest block sizes were chosen randomly

from a uniform distribution between 5 ha and 60 ha based on size ranges common in this region. Harvest blocks were placed preferentially within active “operating areas” and in regions that had high wood volume and road access to

capture the tendency for sets of harvest blocks to be grouped for operational efficiency. An operating area is represented as a circle on the landscape of area 25,000 ha that is active for approximately 20 years, after which time it is deactivated and new operating areas are randomly created within productive forest. Up to five operating areas are active at any given time during a simulation. By preferentially harvesting within operating areas, harvest blocks that are close in time will also tend to be close in space. Simple road construction was also included in the harvest model to account for changing site accessibility through time. Maps of existing roads and projected roads were used to identify the initial road network and simulate future development of main roads. Areas further than 2 km from active roads were not available for harvesting until road access expanded to encompass them. Harvest blocks were explicitly connected to current and future main roads by straight-line, spur road segments. Future roads were activated once a spur road connected to them or when they were required to bridge newly activated segments with the existing road network. This method of modeling road development allowed road access restrictions to influence the spatial harvesting pattern while road-building associated with harvesting reduced access constraints over time.

2.3. Protected areas

PAs were selected to promote the persistence of marten within our boreal forest study area. The area of the PA was fixed at 10–12% (39,200–47,040 ha) of the forested portion of the study

area and could potentially support between 47 (mostly low quality habitat requiring larger home range sizes) and 94 (mostly high quality habitat resulting in smaller home range sizes) marten depending on the density of habitat (percentage cover) within their home ranges. Each PA system was made up of two patches to increase the chance that some portion of the PA would survive in the event of a catastrophic fire and could serve as a source of re-colonization (Shafer, 2001). Without imposing a 2-patch constraint on the number of patches, the PA system would have become highly fragmented and scattered across the landscape in very small patches each comprising only one or a few marten home ranges. This type of PA system would be operationally unrealistic to implement. Furthermore, aggregated PA systems that may include lower quality habitats have been shown to be biologically more valuable than fragmented ones given high levels of uncertainty concerning the effects of fragmentation on species persistence (Moilanen and Wintle, 2006).

2.3.1. Static and dynamic protected area scenarios

For the static PA scenario, both patches of the PA were selected with MARXAN once at the beginning of the 200 year simulation. This static PA was used as the initial PA for the other PA scenarios. For the dynamic PA scenario, both patches were reselected with MARXAN every 50 years during the 200-year simulation (Table 1). We also explored a hybrid of these two scenarios, the static-core scenario, where one patch of the PA was fixed in space for the duration of the simulation while the other was reselected every 50 years. Two baseline

Table 1 – Description of alternative protected area (PA) scenarios

Scenario	Protected area	Logging	Example (Year 0, 50, 100, 150)
No-logging	Entire study area	None	
Static PA	2 static patches	Yes	
Static-core PA	1 static “core” patch, 1 dynamic patch	Yes	
Dynamic PA	2 dynamic patches	Yes	
No-PA	None	Yes	

In the example maps, black represents protected areas (areas where logging is excluded) and white represents areas that are subject to logging. Succession and fire occur across the entire landscape.

scenarios were simulated for reference: a “no-logging” scenario in which only succession and fire were simulated to estimate the historical range of natural variability; and a “no-PA” scenario in which succession, fire, and logging were simulated without designating a PA. Each of the five scenarios was replicated 10 times using our boreal forest simulation model.

2.3.2. Protected area selection method

The PA selection process involved two steps. First, the study area was divided into planning units corresponding to non-overlapping, potential marten home ranges based on the spatial configuration of preferred habitat (Fig. 2a; James et al., 2005). Planning units must be non-overlapping in order to divide up the study area into polygons for potential selection by the reserve selection algorithm. Marten are intrasexually territorial although male home ranges generally overlap with one or more female home ranges depending on habitat conditions and marten densities (Payer et al., 2004), therefore using non-overlapping home ranges provides us with a conservative estimate of marten population size. This home range delineation method assumes that an individual marten may occupy a home range of less than the maximum size provided there is a sufficient density of preferred habitat within that home range (Schultz and Joyce, 1992; Thompson 2004). Preferred marten habitat is mature coniferous and mixed forest (Buskirk and Powell, 1994; Thompson and Harstad, 1994) and at least 50% of each marten home range should consist of this preferred habitat type (Schultz and Joyce, 1992; Potvin et al., 2000). Hence, with a maximum marten home range size of 1000 ha (Potvin et al., 2000), any area up to 1000 ha that contained at least 500 ha of habitat was considered a viable home range. To identify home ranges with a high density of habitat within them, an expanding moving window centered at each cell was applied iteratively across a raster layer of marten preferred habitat. The resulting map of the habitat densities within home ranges (Fig. 1a) was used to prioritize planning units (home ranges) for inclusion in the PA. We used this relatively coarse method of home range delineation to reduce the analysis time and to make best use of the habitat information tracked in the boreal forest simulation model (forest age and composition) while keeping assumptions about space-use and life history at a minimum. This method allows us to go one step beyond simply considering the amount of habitat available to marten in PAs by also incorporating spatial aspects of habitat for species (i.e. the spatial juxtaposition of preferred and non-preferred habitat types) and reflected the variability of marten home range sizes (reviewed in Buskirk and McDonald, 1989).

The second step involved selecting planning units to comprise the PA. We formulated our PA selection problem as a “minimum set covering problem” (Cabeza and Moilanen, 2001): find the set of planning units that minimizes the inclusion of low-density home ranges within a fixed area PA. Formulating our PA selection problem in this way allowed us to employ the existing site-selection software MARXAN (version 1.8.2; (Ball and Possingham, 2000; Possingham et al., 2000)) which uses simulated annealing to approximate a global optimum for large minimum set covering problems. MARXAN has traditionally been used to select a PA that minimizes the area

of the conservation area while meeting fixed targets for the biodiversity surrogate (McDonnell et al., 2002). We simply fixed the target area instead and minimized the difference between the maximum value of the biodiversity surrogate and the actual value in each planning unit (i.e. 100% – actual density of habitat in home range). This was done by modifying the MARXAN input files: the conservation feature file was modified to describe the area-based conservation target for the PA (47,040 ha); the conservation feature distribution file was modified to describe the size of each planning unit; and the planning unit file was modified to describe the difference between the actual density of habitat in the planning unit (home range) and the maximum possible habitat density (100%).

MARXAN solves minimum set covering problems by minimizing a multivariate objective function that gives a value for a collection of planning units as if that collection constituted a PA. Although we modified the objective function, it still consisted of two main sections: the first was a penalty for not meeting the target area (47,040 ha) and the second was a measure of the ‘cost’ of the PA based on the density of habitat within marten home ranges and the boundary length of the entire PA. The objective function we implemented was:

$$\text{total cost} = \text{APF} \times \text{AreaPenalty} + \text{BLM} \sum_{\text{PUs}} \text{BoundaryLengths} \\ + \sum_{\text{PUs}} (100 - \text{HabitatDensity}),$$

where PUs are planning units (marten home ranges), APF is the area penalty factor, a term that assigns a penalty for failing to meet the PA area specified and BLM is the boundary length modifier, a weight used to control the spatial aggregation of selected planning units. The AreaPenalty is a penalty for not adequately meeting the target area for the PA; it is proportionate to the difference between the PA area and the target area. The BLM is used to reduce fragmentation of the PA. If the BLM = 0, the algorithm performs without spatial constraints. Values of the BLM > 0 increase the cost of the boundary length and encourage solutions that aggregate planning units into fewer, compact clusters with shared boundaries. We assigned a value of 10^6 for the APF and adjusted the value for the BLM such that the PA comprised exactly two distinct patches. We used an adaptive annealing schedule with one million iterations and 10,000 ‘temperature’ decreases. The ‘temperature’ is the probability of accepting a change in the set of planning units that comprise the PA even if this change produces a higher objective function value. This probability decreases throughout the algorithm once the chance of prematurely finding a local minimum is reduced. MARXAN was run 10,000 times and we kept the best solution (i.e. the solution that produced the lowest value for the objective function) as the final PA (Fig. 2b).

2.3.3. Comparing protected area scenarios

To compare the PA scenarios (static, static-core, and dynamic) with the baseline scenarios (no-logging and no-PA), we performed 10 replicates of each scenario and monitored two characteristics of marten home ranges both within PAs and across the entire landscape every 50 years: the number of

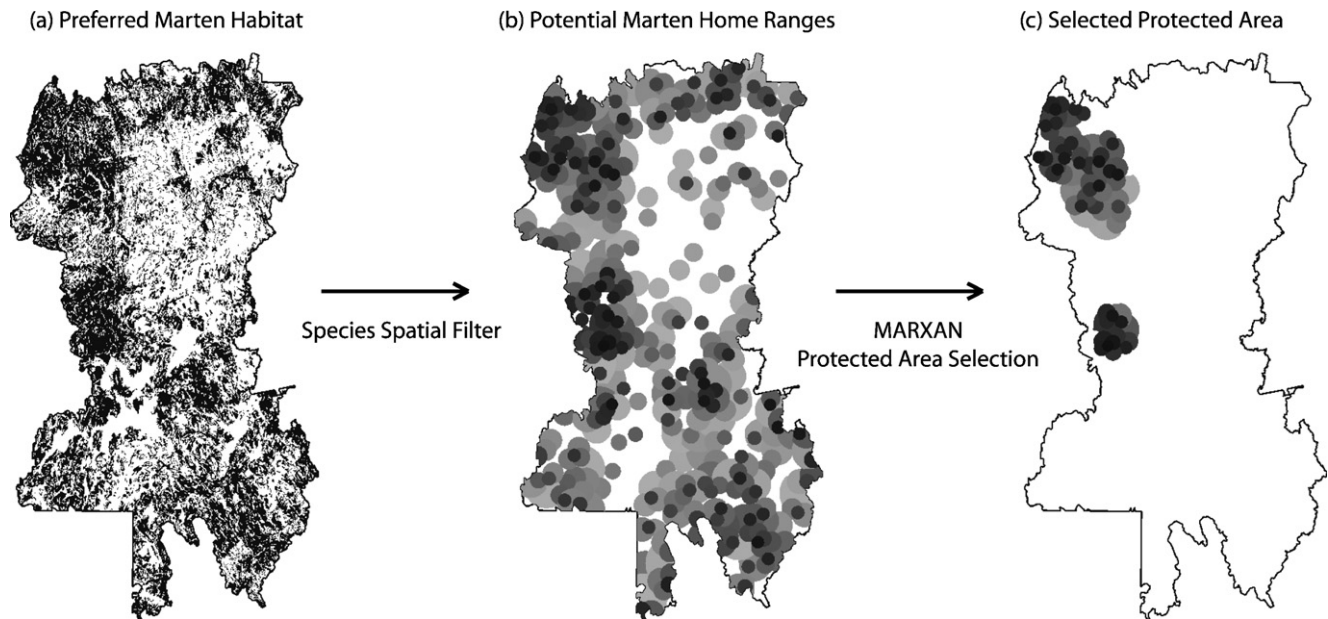


Fig. 2 – Illustration of the two steps of the PA selection process. The first step was to convert potential marten habitat, based on forest composition and age, into potential marten home ranges, based on the spatial configuration of potential habitat and marten movement abilities. The second step was to select potential marten home ranges for inclusion in the PA using the PA selection software MARXAN. MARXAN selected home ranges based on the density of the habitat within them and the boundary length of the PA. Home ranges with high habitat density that were compact were favored for inclusion in the final PA.

home ranges and the mean density of habitat within each home range (home range habitat density). Within PAs we also monitored both the amount of habitat and the amount of area covered by home ranges as a percentage of the area of the PA. To measure home range properties within static PAs (and the core patch of the static-core PAs) we re-ran the home range delineation algorithm strictly within the boundaries of protection. We tracked the boundary length to area ratio of PAs through time using the corrected perimeter-area ratio (Baker and Cai, 1992). This ratio corrects for polygon size by dividing the perimeter of a polygon by the square root of the product of 4π and the area of the polygon. The corrected perimeter-area ratio for a PA was computed as the average of the perimeter-area ratios for both patches within the PA. Its value is always greater than or equal to 1 (with a value of 1.0 for perfect circles) and becomes arbitrarily large for long, skinny polygons.

At the landscape-scale we also monitored home range connectivity (the connectedness among home ranges) measured as the density weighted minimum planar graph (DWMPG; O'Brien et al., 2006). This type of graph-theoretic connectivity analysis summarizes the spatial relationships between landscape elements by building a “graph” (Harary, 1969) consisting of a set of “nodes” that represent landscape elements and “links” that represent the potential movement of an organism among them (Urban and Keitt, 2001). In our particular application the nodes were marten home ranges and the links were Euclidean distances among them. DWMPG is based on a particular type of graph called a minimum planar graph which consists of the maximum number of non-crossing links that connect all nodes (O'Brien et al., 2006). To calculate DWMPG, link lengths are weighted by the density

of habitat in the pair of home ranges that they connect and then a mean is computed for all links in the landscape. Mean DWMPG measures both the structural connectivity and the habitat density (quality) of connected home ranges. Its value is large when home ranges are far apart and have low habitat densities, whereas it is small when home ranges are close together and have high habitat densities.

3. Results

None of the PA scenarios (static, static-core, and dynamic) dramatically outperformed the others based on any of the measured properties of marten home ranges across the entire landscape over the 200 years of simulation: number (Fig. 3a), habitat density (Fig. 3b), or connectivity (Fig. 3c). Error bars representing $\pm$ two standard errors were highly overlapping among the three scenarios at this landscape-scale (Fig. 3). However, the dynamic and static-core PA scenarios performed at least as well if not slightly better than the static PA scenario at all time steps for all home range properties. Not surprisingly, all three PA scenarios were always worse than the no-logging scenario and better than the no-PA scenario. The number of home ranges across the entire landscape decreased initially but then remained fairly constant through time for all PA scenarios (Fig. 3a). The density of habitat within home ranges across the entire landscape also remained fairly constant for all scenarios except for the no-logging scenario, which improved gradually, and the no-PA scenario, which declined (Fig. 3b). The dynamic PA scenario produced the most consistent home range habitat density across the landscape through time; however, even for the

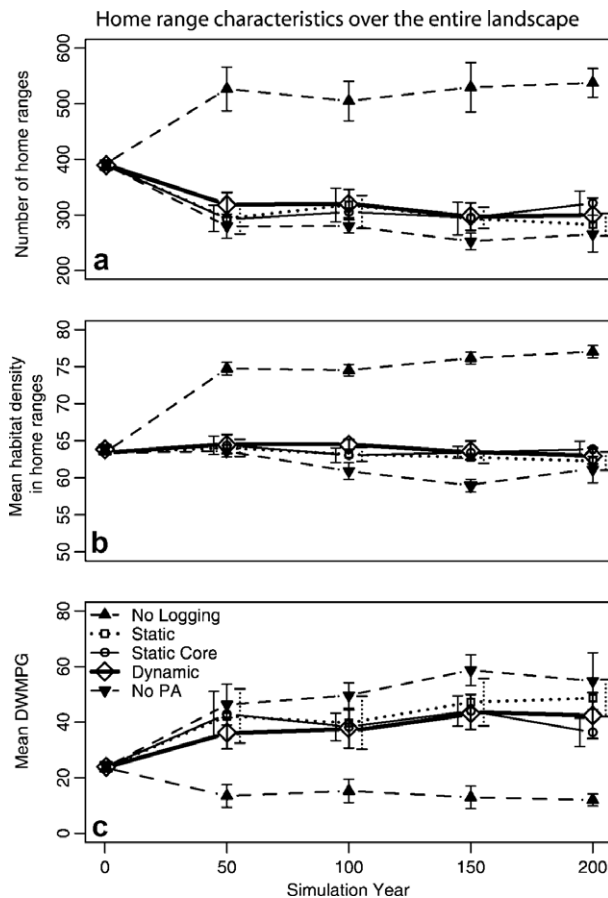


Fig. 3 – Mean properties of all home ranges in the study area (both inside and outside of the PAs) over 10 replicates for the alternative PA scenarios at initial conditions and at 50, 100, 150 and 200 years of simulation: (a) number of home ranges; (b) mean density of habitat within home ranges and (c) mean density-weighted minimum planar graph (DWMPG). Error bars represent ± 2 SE and are offset to the left (static-core scenario) and right (static scenario) of mean values to facilitate visualization.

static PA scenario, mean home range habitat density only ranged from 67.88% to 72.80%. The connectedness among all home ranges in the landscape decreased for all three PA scenarios, as illustrated by the increase in mean DWMPG (Fig. 3c). The dynamic PAs maintained connectivity better than the static and static-core PAs for the first 150 years of simulation. The dynamic and static-core PAs resulted in the highest levels of connectedness among home ranges after the full 200 years of simulation and differences among the three scenarios were most pronounced at this final time step; however, these differences were still small enough and error bars were large enough that they may have been due to stochasticity alone.

We also tracked home range properties strictly within the PAs and found stronger evidence of the dynamic and static-core PA scenarios outperforming the static PA scenario (Fig. 4). We standardized all PA-scale analyses by the area of the PA, which fluctuated between 10.34% and 12.13% of the forested portion of the study area. The percentage of PAs that contained habitat was higher in the dynamic PAs than the

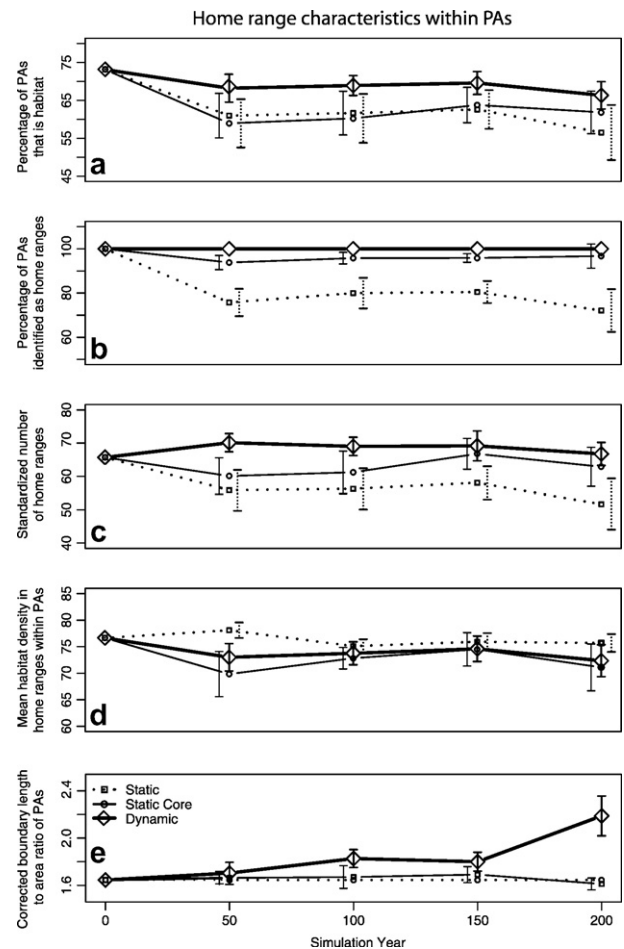


Fig. 4 – Mean properties of PAs over 10 replicates for the static, static-core, and dynamic PA scenarios at initial conditions and at 50, 100, 150 and 200 years of simulation: (a) percentage of PAs that is habitat (i.e. the area of habitat within each PA divided by the total area of the PA); (b) Percentage of PAs identified as home ranges (i.e. the cumulative area of home ranges within each PA divided by the total area of the PA); (c) Standardized number of home ranges (i.e. the number of home ranges within each PA divided by the maximum possible home ranges of 100% habitat density that would fit in the area of the PA); (d) Mean density of habitat within home ranges in the PAs and (e) Corrected boundary length to area ratio of PAs (see text for details). Error bars represent ± 2 SE and are offset to the left (static-core scenario) and right (static scenario) of mean values to facilitate visualization.

static-core or static PAs and was more consistent through time (Fig. 4a). Differences among scenarios were larger when considering either the percentage of PAs that contained home ranges (Fig. 4b) or the number of home ranges in PAs (Fig. 4c) rather than simply the amount of habitat. The dynamic and static-core PAs maintained significantly higher percentages of home ranges than the static PAs over time because they were re-selected by MARXAN at the time of analysis while the percentage of home ranges in static PAs declined due to stochastic losses from fire (Fig. 4b). The number of home ranges in the dynamic PAs was also consistently higher than

the static PAs but the static-core PAs were more variable (Fig. 4c). In contrast, habitat density of home ranges within PAs was best maintained by the static scenario; however, after 100 years mean values for all scenarios were very similar and error bars among all scenarios were highly overlapping (Fig. 4d). Variability in the habitat density of home ranges was highest in the static-core scenario, which ranged from 58% to 82%, and lowest in the static scenario, which only ranged from 72% to 81%. Taken together, Fig. 4b–d show that home ranges in static PAs were less numerous and cumulatively covered a smaller area of the PAs but had similar if not slightly higher habitat density than home ranges in the static-core and dynamic PAs.

The boundary length to area ratio of the static, static-core, and dynamic PAs showed a different pattern (Fig. 4e). The ratio was constant for the static PA because both patches in the static PA were fixed in space. Similarly, the static-core PA scenario maintained a consistent ratio largely because the core patch (which was fixed in space) was very compact and nearly circular. In contrast, the ratio sharply increased for the dynamic PAs, in the final time step suggesting that high quality home ranges became increasingly discontinuous in the dynamic PA scenario.

4. Discussion

In this paper we compared the performance of static and dynamic PAs in a dynamic forest landscape based on their abilities to maintain through time important habitat characteristics of a model species, the American marten, as it is an indicator of mature forest. Dynamic PAs maintained more home ranges than static PAs when implemented within our spatially-explicit boreal forest dynamics model with a clear cut logging regime and an annual allowable cut of 1%. Home ranges within dynamic PAs were of comparable habitat density to home ranges in static PAs even though they were more numerous. The benefits of dynamic PAs were only observable when home range characteristics were considered within PAs and were not seen at the landscape-scale. All three PA scenarios were much more similar to the no-PA scenario than the no-logging scenario at the landscape-scale likely due to the area constraint on PAs of 10–12% of the forested area. These general patterns can be attributed to the PA selection methods because the annual allowable cut of the logging regime was kept constant in all scenarios (i.e. the logging rate was not decreased when a portion of the area was protected). The performance of dynamic PAs was constrained by the fragmentation of forest in unprotected areas due to interactions between fire and harvest, which limited options for re-locating PAs during the 200 year simulations. The increasing boundary length to area ratio of the dynamic PAs through time reflected a reduction in the contiguity of high quality home ranges remaining on the landscape.

Several conceptual papers have suggested the use of dynamic PAs in boreal forest ecosystems (Cumming et al., 1996; Bengtsson et al., 2003; Angelstam et al., 2004); however, they have not discussed the details of their implementation. Our study is the first to have fully implemented a dynamic PA by selecting PAs every 50 years with MARXAN and simulat-

ing landscape change in a boreal forest with a spatially-explicit stochastic landscape model. Our results are undoubtedly influenced by the way in which we defined focal species habitat, implemented the dynamic PAs, represented landscape processes and captured interactions between the disturbance regime and PA identification. Two critical implementation parameters likely affected the performance of dynamic PAs: the length of the re-planning interval and the logging regime applied in unprotected areas. The longer the interval between re-planning, the larger effect logging will have on unprotected areas which will in turn limit continuous tracks of forest that could serve as options for re-locating PAs. However, if the re-planning interval is too short, then the probability of a PA burning would be too small to merit the effort required to re-plan and re-locate the PA. Therefore, an intermediate re-planning interval informed by both the logging and fire regimes is desirable.

This problem of choosing a re-planning interval is also of interest to studies examining the periodic acquisition of PAs over an implementation horizon comprising several time periods (Costello and Polasky, 2004; Meir et al., 2004; Snyder et al., 2004; Drechsler, 2005; Rush O'Hanley et al., 2007). These studies recognize that budget constraints force PAs to be implemented over several planning periods and that uncertainty in site availability and habitat loss during the implementation horizon may compromise the effectiveness and efficiency of the PAs selected. They have considered a range in the number of planning periods from two to ten (Costello and Polasky, 2004; Snyder et al., 2004; Drechsler, 2005; Strange et al., 2006) without specifying the length of each planning period (with the exception of Meir et al. (2004) who recommended an annual re-planning interval over a 10 year planning horizon). Although the length of the planning periods would affect the probability of site availability in these studies, they have placed greater concern on the number of planning periods because adding more planning periods increases the computational load. Oetting et al. (2006) implemented a dynamic PA selection process by updating their broad scale land acquisition plan in Florida at six month intervals during a 10 year implementation horizon in response to land use changes and conservation actions. Strange et al. (2006) showed that allowing planners to swap PAs that become degraded through time increases the overall efficiency of the PA system in terms of the number of species represented for a given budget. However, they did not specify the length of the re-planning interval and they assumed that degradation of PAs was irreversible and that PA selection was not spatially-explicit. Given that we simulated boreal forest dynamics over 200 years and selected spatially-explicit PAs using traditional single planning period site-selection methods, the planning periods used in these studies do not directly apply. In our system, a “dynamic” re-planning period may be appropriate, in which re-planning only occurs when the conservation value of a PA has dropped below some threshold (e.g. due to fires). We identify the issue of re-planning intervals as an area that needs more research, particularly as the dynamics of ecological and human systems are increasingly acknowledged as important drivers for the persistence of biodiversity in PAs.

Another implementation parameter that likely affected the performance of the dynamic PAs was the type of logging

regimes implemented in unprotected areas. It is becoming increasingly recognized that conservation of forest biodiversity is not only dependent upon protected PAs, but also the management of the forest matrix surrounding them (Lindenmayer et al., 2006). Studies examining the consequences of destroying habitat that is not selected for inclusion in a PA have shown that the probability of occurrence of the species inside the PA is reduced when all habitat outside PAs is lost (Cabeza, 2003; Cabeza and Moilanen, 2003). In the case of multi-use boreal forests however, habitat outside PAs is not completely lost due to the arrangement of logging activities in space and time. The spatial and temporal distribution of logging can be managed to promote mature forest habitat retention in the unprotected areas of the landscape (Lindenmayer et al., 2006). Designing alternative logging regimes may involve modifying three variables (Bergeron et al., 1999): size of the harvest unit, internal composition of the harvest unit, or period of time between successive logging events in a harvest unit (i.e. rotation length). Imposing a spatial constraint on the minimum harvest unit size would help to reduce fine-scale fragmentation and if harvesting occurred in patterns more in line with marten home ranges, then it may be easier to find new good protected areas during re-planning. Modifying the composition of harvest units under the clear-cutting system of even-aged stand management would help maintain more structural complexity in the form of snags and coarse woody debris within harvest units. Yet, the internal structural heterogeneity of harvest units would not affect our assessments of dynamic PAs as it is at a finer scale than we have modeled here.

The rotation length determines the rate at which the landscape is disturbed and consequently the proportion of mature forest on the landscape. Longer rotation lengths would increase the options for re-location of the PAs to mature forest areas, however it would also reduce the volume of timber flowing from the forest (Bergeron et al., 2001). Another option is to protect a “training PA” that is a large tract of land that will be suitable mature forest habitat during the next re-planning period even if it is not suitable during the current planning interval. This option suggests the need to conserve an area larger than “needed” to ensure sustainable “recruitment” of high quality PAs over time. This option may reduce negative effects on the logging industry while increasing the probability of a high quality option for re-location of the dynamic PA.

Initial conditions in our study were created by real interactions among management and fire and represent a realistic set of spatial constraints with regards to forest age and species composition in this region of the boreal forest. Real options for management in general, and for PA selection specifically, are necessarily constrained by the history of disturbance and management within a study area (Gustafson, 1998) and will also depend on current management and anticipated future changes in disturbance. Although the results of our study are specific to the study region, disturbance regime, and stochastic model used, the general qualitative messages emerging from these analyses are broadly applicable and should be instructive for many different systems. One important message that can be extracted from this localized case study is that small, dynamic PAs will not necessarily solve the problem of our current unwillingness to set aside large

PAs that can accommodate landscape dynamics within their boundaries (Pickett and Thompson, 1978). The ability of dynamic PAs to safeguard against further losses of biodiversity will undoubtedly hinge on the details of their implementation and a key component of their success will be to manage the matrix surrounding PAs in such a way that options remain when it comes to re-planning.

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